

Grade 12 Maths Unit 3

Introduction

Competencies

- know absolute and relative dispersion and their interpretation.
- deduce specific facts about measurement in statistical data.

3.1 Measure of Absolute Dispersion

Competencies

- compute the range of a given data.
- describe the relative significance of range as a measure for dispersion.
- calculate inter-quartile range of a given data set.
- describe the relative significance of inter-quartile range as a measure for dispersion.
- compute mean deviation of a given data set.
- describe the relative significance of mean deviation as a measure for dispersion.
- calculate the quartile deviation of a given data set.
- describe the relative significance of quartile deviation as a measure for dispersion.
- find the variance of a given data set.
- describe the relative significance of variance as a measure of dispersion.
- calculate the standard deviation of a given data.
- describe the usefulness of standard deviation in interpreting the variability of a given data.

3.2 Interpretation of Relative Dispersion

Competencies

- calculate coefficient of range of a given data set.
- interpretation of two or more data by using their coefficient ranges.
- determine the coefficient of quartile deviation of a given data set.
- interpret two or more data sets by using their coefficient of quartile deviation.
- calculate the coefficient of mean deviation of a given data set.
- interpret two or more data sets by using their coefficient of mean deviation.
- calculate the coefficient of variation of a given data set.

3.3 Use of Frequency Curves

Competencies

- describe the relationship between mean, median and mode for grouped data using its frequency curve.
- use cumulative frequency graphs to determine the dispersion of values of data (in terms of mean, median and mode).
- determine the variability of data values in terms of quartiles by using a cumulative frequency graph.

3.4 Sampling Techniques

Competencies

By the end of this unit, students will be able to:

- describe the four methods/ techniques of sampling.
- explain the advantages and limitations of each sampling technique.